ADVERSARIAL SIGNALS

Linguistic Fragility in Earnings Calls — Equity Underperformance in Stressed US Firms

[Author Name] · [University] · April 2026 · SSRN Working Paper

RESEARCH SUMMARY

NLP · Asset Pricing · Systematic

31,030 obs · 566 firms · 2005–2023

−2.76	−3.46	−4.67	63 bps	+0.61
FM t-stat	**LFS+**	**Mom Loser t**	**Net Alpha**	**Placebo t**
p < .01 | LFS	*FM t | Enhanced*	*Q5−Q1 = −2.36%*	*After 30% TO costs*	*Non-spurious ✓*

HYPOTHESIS

Markets systematically underweight adversarial linguistic signals embedded in earnings call Q&A, especially for firms already under fundamental stress. This underreaction generates predictable return patterns over 1–6 month horizons.

LINGUISTIC FRAGILITY SCORE (LFS)

```
LFS = 0.30×z_uncertainty + 0.25×z_evasiveness
+ 0.25×z_shift + 0.10×z_filler −
0.10×z_numeric
```

- Within-firm rolling z-scores (8Q window) — removes style bias
- Q&A evasiveness: cosine(question, response) embedding
- PCA (t=−3.76) and Lasso (t=−3.60) confirm signal without human-chosen weights

TWO-STAGE INCORPORATION

FM t-statistic by forward return horizon:

1 month (Stage 1: partial): **t = −4.19** *Immediate sophisticated-investor reaction*

3 months (transition): **t = −1.34** *Gradual analyst revision processing*

6 months (Stage 2: full diffusion): **t = −3.83** *Primary specification; full incorporation*

12 months (resolved): **t = +0.22** *Signal fully priced in*

PORTFOLIO RESULTS

Strategy	
LFS Q5−Q1 (raw)	
Sector-Neutral	
Beta-Neutral	
LFS ∩ Mom Loser 🔥	

3×3 DOUBLE SORT: MOMENTUM × LFS

MOM Group	L
Losers (bottom 33%)	+2.31%
Middle (mid 33%)	+2.33%
Winners (top 33%)	+1.24%

Sign reversal in Winners confirms LFS is a conditional enhancer, not a standalone factor.

OUT-OF-SAMPLE VALIDATION

Period	
In-Sample (2005–2014)	
Out-of-Sample (2015–2023)	

Portfolio return stable OOS (−1.00% vs −0.98% IS). IC decline reflects high per-stock noise, not signal failure.

TECH STACK

Python 3.12 · SBERT all-mpnet-base-v2 · Lasso / Ridge / RF / GBM · Fama-MacBeth · Factor sorts

DATA

Refinitiv Streetevents (WRDS) · CRSP · Compustat · Fama-French + Momentum factors

HONEST LIMITS

Extended orth reduces t: −2.76 → −1.58. 6F alpha t = −1.83 (marginal). Signal decays 62.4% by 2022.

CONTACT

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